Mr Michael Weiss

Operating System

Proof of Concept Linux virtual network Project

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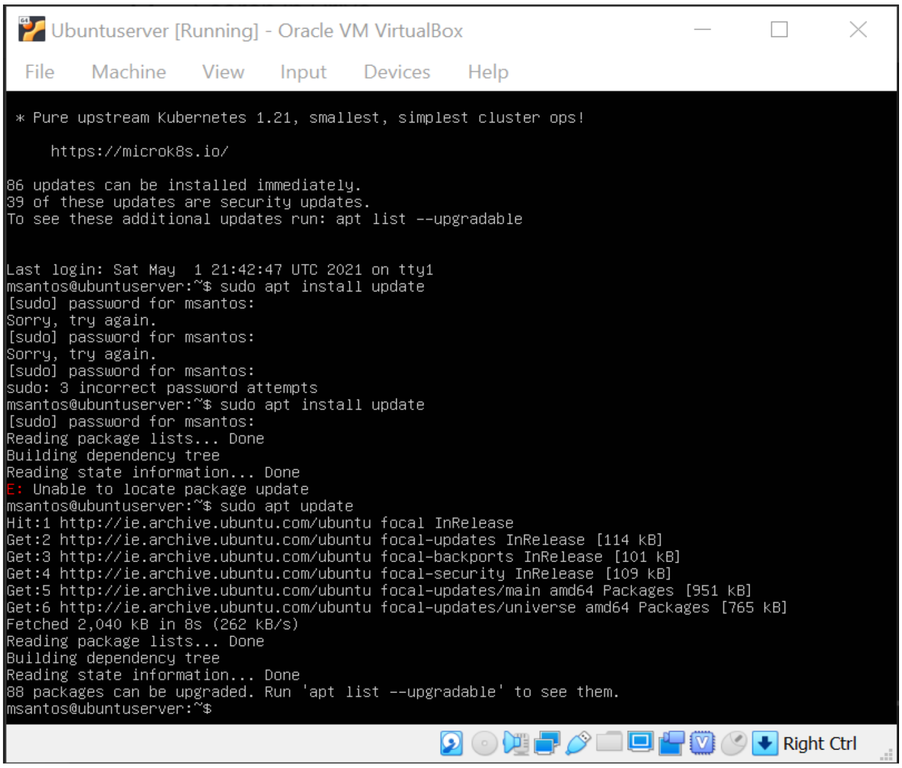
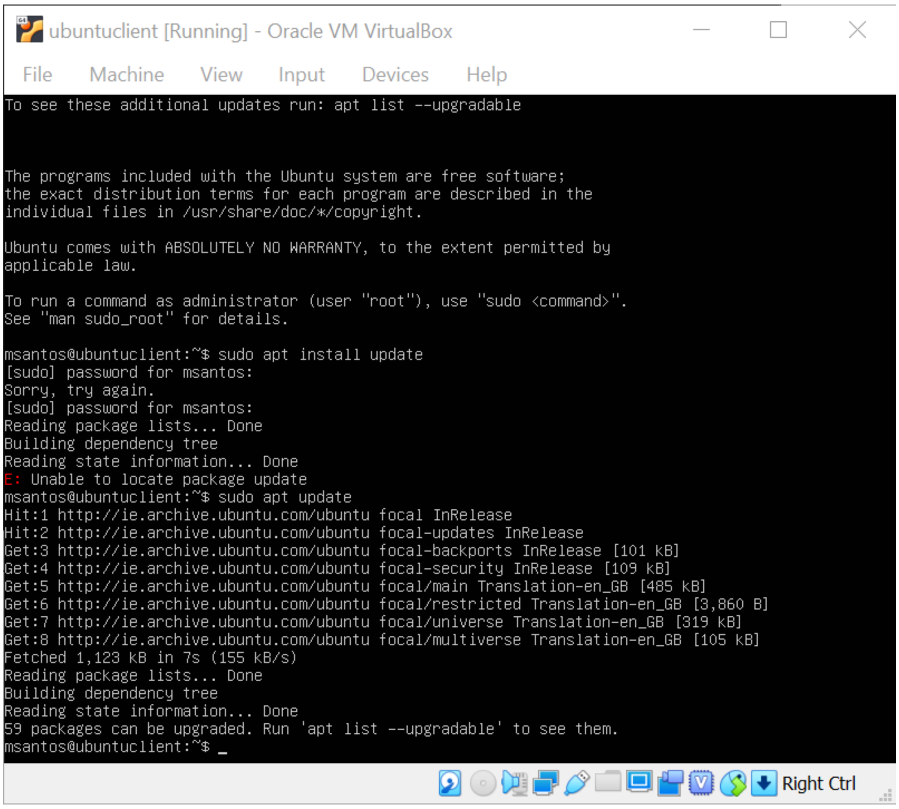
Summary

Obtain Linux updated and ping to the servers

Ubuntuserver update



Ubuntuclient update



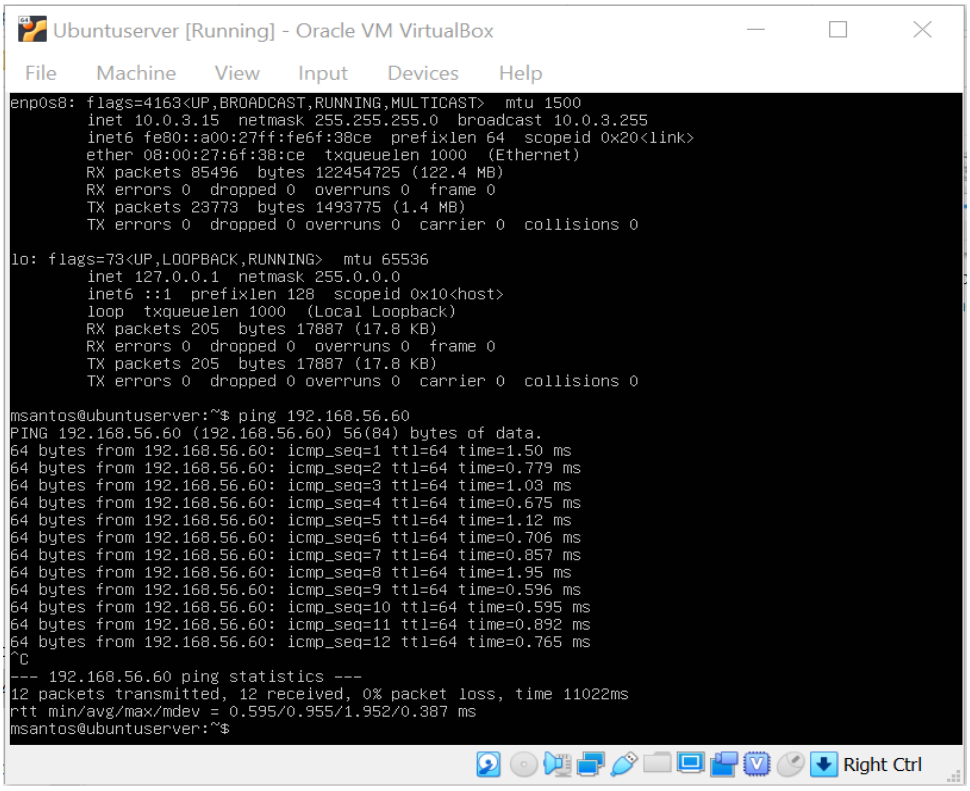
Ubuntuserver upgrade

```
E: Unable to locate package upgrade
msantos@ubuntuserver:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libevdev2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssh-server openssh-sftp-server openssl pciutils pollinate
  procps python3-apport python3-distupgrade python3-problem-report python3-software-properties
  python3-twisted python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen
  snapd software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
96 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 248 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

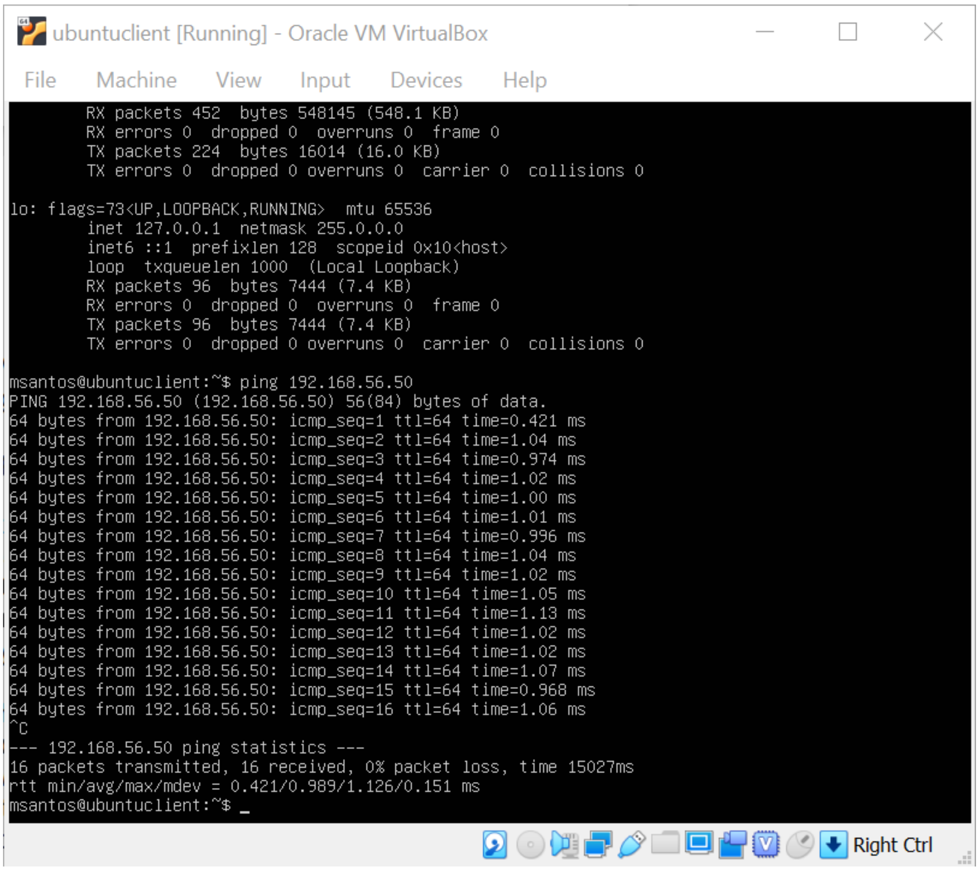
Ubuntuclient upgrade

```
E: Unable to locate package upgrade
msantos@ubuntuclient:~$ sudo apt upgrade
Reading package lists... Done
Building dependency tree
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  linux-headers-5.4.0-65 linux-headers-5.4.0-65-generic linux-image-5.4.0-65-generic
  linux-modules-5.4.0-65-generic linux-modules-extra-5.4.0-65-generic
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  libevdev2 libimobiledevice6 libplist3 libupower-glib3 libusbmuxd6 linux-headers-5.4.0-72
  linux-headers-5.4.0-72-generic linux-image-5.4.0-72-generic linux-modules-5.4.0-72-generic
  linux-modules-extra-5.4.0-72-generic upower usbmuxd
The following packages will be upgraded:
  alsa-ucm-conf apport apt apt-utils bind9-dnsutils bind9-host bind9-libs ca-certificates
  cloud-init curl dirmngr distro-info-data friendly-recovery git git-man gnupg gnupg-l10n
  gnupg-utils gpg gpg-agent gpg-wks-client gpg-wks-server gpgconf gpgsm gpgv grub-common grub-pc
  grub-pc-bin grub2-common initramfs-tools initramfs-tools-bin initramfs-tools-core
  isc-dhcp-client isc-dhcp-common landscape-common libapt-pkg6.0 libasound2 libasound2-data
  libcurl3-gnutls libcurl4 libglib2.0-0 libglib2.0-bin libglib2.0-data libhogweed5 libldap-2.4-2
  libldap-common libnettle7 libnss-systemd libpam-systemd libpci3 libprocps8 libpython3.8
  libpython3.8-minimal libpython3.8-stdlib libseccomp2 libssl1.1 libsystemd0 libudev1 libxmlb1
  libzstd1 linux-firmware linux-generic linux-headers-generic linux-image-generic open-iscsi
  open-vm-tools openssh-client openssl pciutils pollinate procps python3-apport
  python3-distupgrade python3-problem-report python3-software-properties python3-twisted
  python3-twisted-bin python3-update-manager python3.8 python3.8-minimal screen snapd
  software-properties-common sosreport systemd systemd-sysv systemd-timesyncd thermald tmux
  ubuntu-keyring ubuntu-release-upgrader-core udev update-manager-core update-notifier-common
94 upgraded, 12 newly installed, 0 to remove and 0 not upgraded.
Need to get 247 MB of archives.
After this operation, 382 MB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

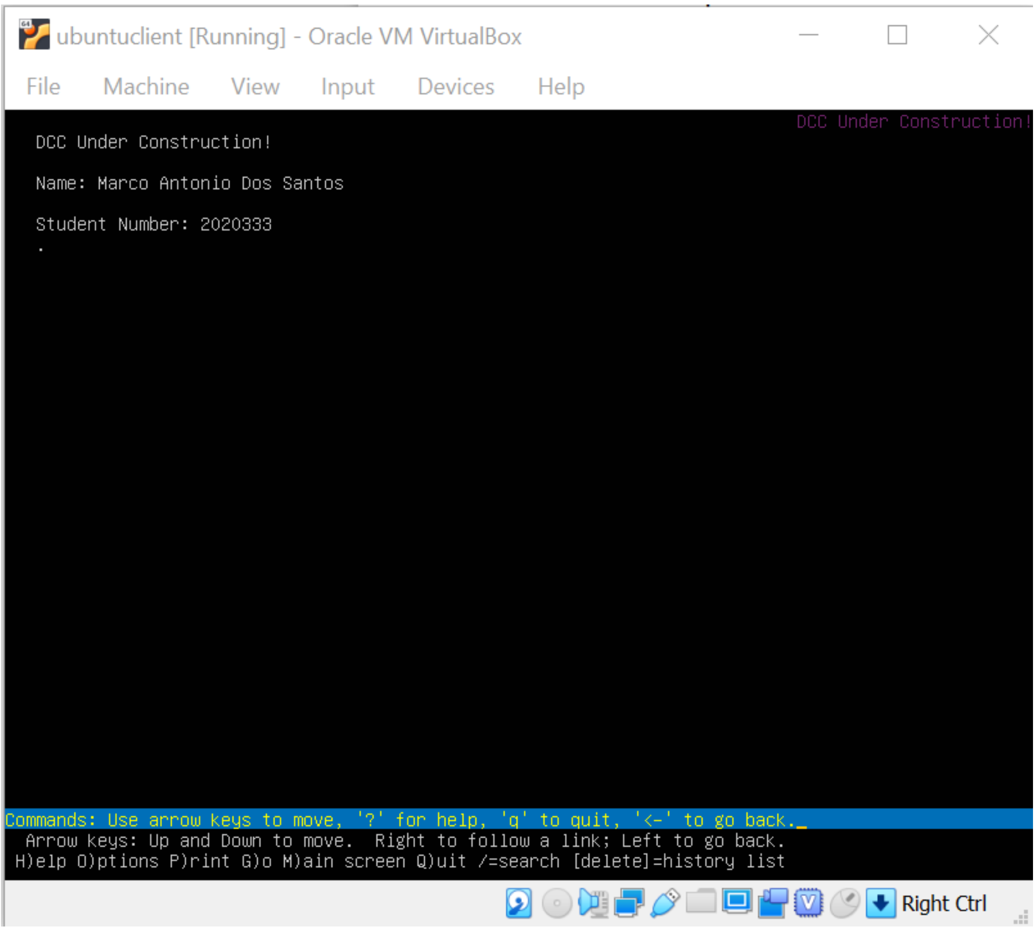
UbuntuServer pinging to the UbuntuClient



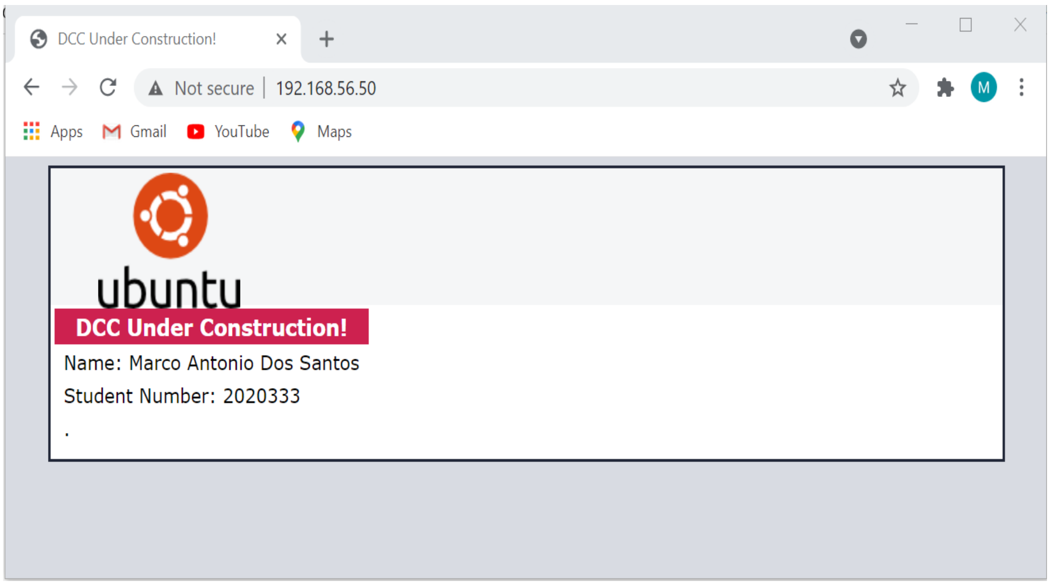
UbuntuClient pinging to the UbuntuServer



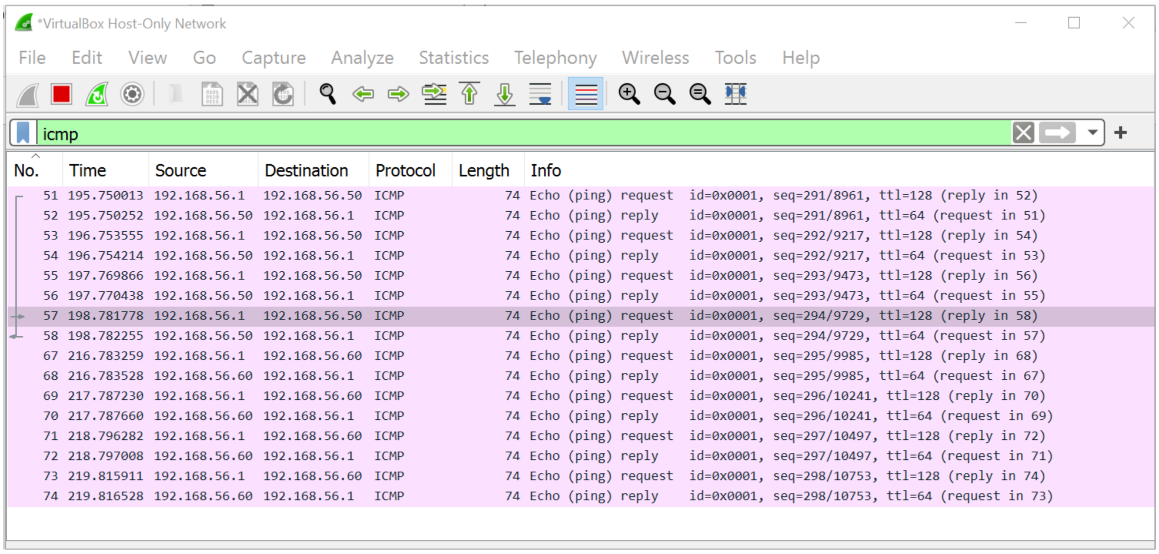
Test access to Apache home page!



Host operating system to access to the Apache home page

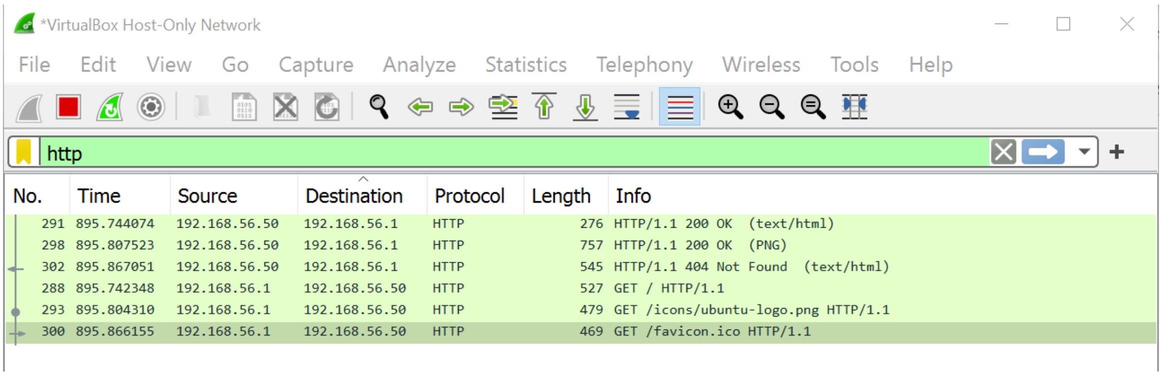


Wireshark capture of Icmp traffic



No.	Time	Source	Destination	Protocol	Length	Info
51	195.750013	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=291/8961, ttl=128 (reply in 52)
52	195.750252	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=291/8961, ttl=64 (request in 51)
53	196.753555	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=292/9217, ttl=128 (reply in 54)
54	196.754214	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=292/9217, ttl=64 (request in 53)
55	197.769866	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=293/9473, ttl=128 (reply in 56)
56	197.770438	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=293/9473, ttl=64 (request in 55)
57	198.781778	192.168.56.1	192.168.56.50	ICMP	74	Echo (ping) request id=0x0001, seq=294/9729, ttl=128 (reply in 58)
58	198.782255	192.168.56.50	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=294/9729, ttl=64 (request in 57)
67	216.783259	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=295/9985, ttl=128 (reply in 68)
68	216.783528	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=295/9985, ttl=64 (request in 67)
69	217.787230	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=296/10241, ttl=128 (reply in 70)
70	217.787660	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=296/10241, ttl=64 (request in 69)
71	218.796282	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=297/10497, ttl=128 (reply in 72)
72	218.797008	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=297/10497, ttl=64 (request in 71)
73	219.815911	192.168.56.1	192.168.56.60	ICMP	74	Echo (ping) request id=0x0001, seq=298/10753, ttl=128 (reply in 74)
74	219.816528	192.168.56.60	192.168.56.1	ICMP	74	Echo (ping) reply id=0x0001, seq=298/10753, ttl=64 (request in 73)

HTTP contents being transferred between servers



No.	Time	Source	Destination	Protocol	Length	Info
291	895.744074	192.168.56.50	192.168.56.1	HTTP	276	HTTP/1.1 200 OK (text/html)
298	895.807523	192.168.56.50	192.168.56.1	HTTP	757	HTTP/1.1 200 OK (PNG)
302	895.867051	192.168.56.50	192.168.56.1	HTTP	545	HTTP/1.1 404 Not Found (text/html)
288	895.742348	192.168.56.1	192.168.56.50	HTTP	527	GET / HTTP/1.1
293	895.804310	192.168.56.1	192.168.56.50	HTTP	479	GET /icons/ubuntu-logo.png HTTP/1.1
300	895.866155	192.168.56.1	192.168.56.50	HTTP	469	GET /favicon.ico HTTP/1.1

2- SSH logging and encrypted traffic

Logging into ubuntuserver using SSH

```
msantos@ubuntuserver: ~
login as: msantos
msantos@192.168.56.50's password:
Welcome to Ubuntu 20.04.2 LTS (GNU/Linux 5.4.0-65-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

  System information as of Sat  8 May 11:34:40 UTC 2021

  System load:  0.01              Processes:               116
  Usage of /:   50.0% of 8.79GB   Users logged in:         1
  Memory usage: 24%               IPv4 address for enp0s3: 192.168.56.50
  Swap usage:   0%                IPv4 address for enp0s8: 10.0.3.15

 * Pure upstream Kubernetes 1.21, smallest, simplest cluster ops!

     https://microk8s.io/



*** System restart required ***
Last login: Sat May  8 11:32:24 2021 from 192.168.56.1
msantos@ubuntuserver:~$
```

SSH encrypted Traffic

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.56.50	192.168.56.1	SSH	182	Server: Encrypted packet (len=128)
2	0.040843	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=129 Win=8209 Len=0
3	1.014683	192.168.56.50	192.168.56.1	SSH	166	Server: Encrypted packet (len=112)
4	1.055102	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=241 Win=8209 Len=0
5	2.016412	192.168.56.50	192.168.56.1	SSH	166	Server: Encrypted packet (len=112)
6	2.056893	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=353 Win=8208 Len=0
7	2.373791	192.168.56.1	239.255.255.250	SSDP	171	M-SEARCH * HTTP/1.1
8	2.395066	192.168.56.1	239.255.255.250	SSDP	166	M-SEARCH * HTTP/1.1
9	3.017464	192.168.56.50	192.168.56.1	SSH	182	Server: Encrypted packet (len=128)
10	3.057231	192.168.56.1	192.168.56.50	TCP	54	52596 → 22 [ACK] Seq=1 Ack=481 Win=8208 Len=0
11	4.022642	192.168.56.50	192.168.56.1	SSH	182	Server: Encrypted packet (len=128)

3-Rename Hostname(s)

Ubuntuservers rename

```
root@ubuntuserver:/home/msantos# cat /etc/hostname
ubuntuserver
root@ubuntuserver:/home/msantos# hostnamectl set-hostname web-server-333
root@ubuntuserver:/home/msantos# hostname
web-server-333
root@ubuntuserver:/home/msantos# hostnamectl
   Static hostname: web-server-333
         Icon name: computer-vm
           Chassis: vm
        Machine ID: 40621680576c401a8827ef4d7fbeab67
           Boot ID: f0ca8fbbb88f4b2193c1abc871dfe6c0
    Virtualization: oracle
  Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-65-generic
      Architecture: x86-64
root@ubuntuserver:/home/msantos# hostname
web-server-333
```

Ubuntuclient rename

```
root@ubuntuclient:/home/msantos# cat /etc/hostname
ubuntuclient
root@ubuntuclient:/home/msantos# hostnamectl set-hostname web-client-333
root@ubuntuclient:/home/msantos# hostname
web-client-333
root@ubuntuclient:/home/msantos# hostnamectl
   Static hostname: web-client-333
         Icon name: computer-vm
           Chassis: vm
        Machine ID: 9a0687241d3e4aecadb9047e9e74d4cc
           Boot ID: b87e49cf7ae84c60ab1eaa98ae091950
    Virtualization: oracle
  Operating System: Ubuntu 20.04.2 LTS
            Kernel: Linux 5.4.0-72-generic
      Architecture: x86-64
root@ubuntuclient:/home/msantos# hostname
web-client-333
```

Setting the Ip address permanent

The directory and file was created but unfortunately it was not possible to set ip address permanent of both Ubuntu servers ,look at images below:

Ubuntuserver:

```
  GNU nano 4.8              /etc/netplan/00-installer-config.yaml              Modified
# This is the network config written by 'subiquity'
network:
 version: 2
 renderer: networkd
 ethernets:
  enp0s3:
    dhcp4: no
    dhcp6: no
    addresses:[192.168.56.100/24]
    gateway4: 192.168.56.1
    nameservers:
     addresses:[8.8.8.8, 1.1.1.1]
```

```
msantos@web-server-333:~$ sudo netplan try
Error while loading /etc/netplan/00-installer-config.yaml, aborting.
msantos@web-server-333:~$ sudo netplan apply
/etc/netplan/00-installer-config.yaml:10:5: Invalid YAML: inconsistent indentation:
    gateway4: 192.168.56.1
    ^
msantos@web-server-333:~$ _
```

As you can see in this ubuntuserver tell us a error in gateway4.

Ubuntuclient

```
  GNU nano 4.8                /etc/netplan/00-installer-config.yaml                Modified
# This is the network config written by 'subiquity'
network:
 version: 2
 renderer: networkd
 ethernets:
  enp0s3:
    dhcp4: no
    dhcp8: no
    addresses: [192.168.56.200/24]
    gateway4: 192.168.56.1
    nameservers:
     addresses: [8.8.8.8, 1.1.1.1]
```

```
msantos@web-client-333:~$ sudo netplan apply
/etc/netplan/00-installer-config.yaml:8:5: Error in network definition: unknown key 'dhcp8'
    dhcp8: no
    ^

msantos@web-client-333:~$ sudo netplan try
/etc/netplan/00-installer-config.yaml:8:5: Error in network definition: unknown key 'dhcp8'
    dhcp8: no
    ^

An error occurred: the configuration could not be generated

Reverting.
Warning: Stopping systemd-networkd.service, but it can still be activated by:
  systemd-networkd.socket
msantos@web-client-333:~$
```

In these Ubuntuserver tell us that error in dhcp8, both server were using the same proceeding however gave a few different errors, maybe they were caused by spaces errors or even typing.

4- Firewall

SSH not permitted

```
msantos@web-server-333:~$ sudo ufw deny in ssh
[sudo] password for msantos:
Skipping adding existing rule
Skipping adding existing rule (v6)
msantos@web-server-333:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     DENY        Anywhere
22/tcp (v6)                DENY        Anywhere (v6)
```

HTTP traffic not being permitted

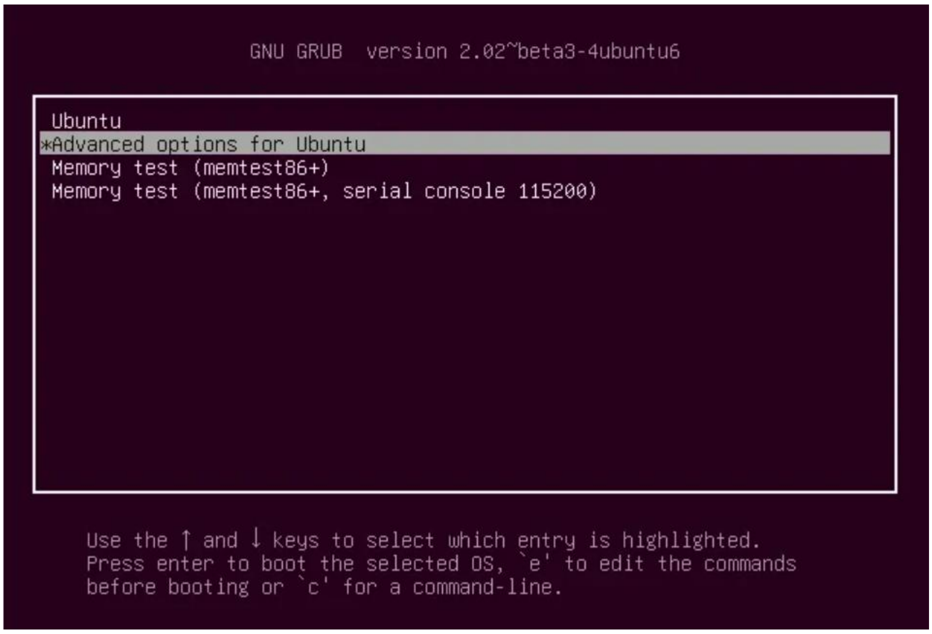
```
msantos@web-server-333:~$ sudo ufw deny out http
Rule added
Rule added (v6)
msantos@web-server-333:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     DENY        Anywhere
80/tcp                     DENY        Anywhere
22/tcp (v6)                DENY        Anywhere (v6)
80/tcp (v6)                DENY        Anywhere (v6)

80/tcp                     DENY OUT    Anywhere
80/tcp (v6)                DENY OUT    Anywhere (v6)

msantos@web-server-333:~$
```

SSH being allow and other traffic not permitted

```
msantos@web-server-333:~$ sudo ufw status
Status: active

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW       Anywhere
80/tcp                     DENY        Anywhere
22/tcp (v6)                ALLOW       Anywhere (v6)
80/tcp (v6)                DENY        Anywhere (v6)

80/tcp                     DENY OUT    Anywhere
80                         DENY OUT    Anywhere
80/tcp (v6)                DENY OUT    Anywhere (v6)
80 (v6)                    DENY OUT    Anywhere (v6)

msantos@web-server-333:~$ _
```

5-How to recover the Password Of Ubuntu Linux

It's possible easily redefine reset a user's password in Ubuntu using a few commands at system startup. It must initiate Ubuntu machine and press Esc Key continuously after the bios screen to open the Ubuntu grub menu , as shown below. Then, select "Advanced option for Ubuntu" and press the Enter key.

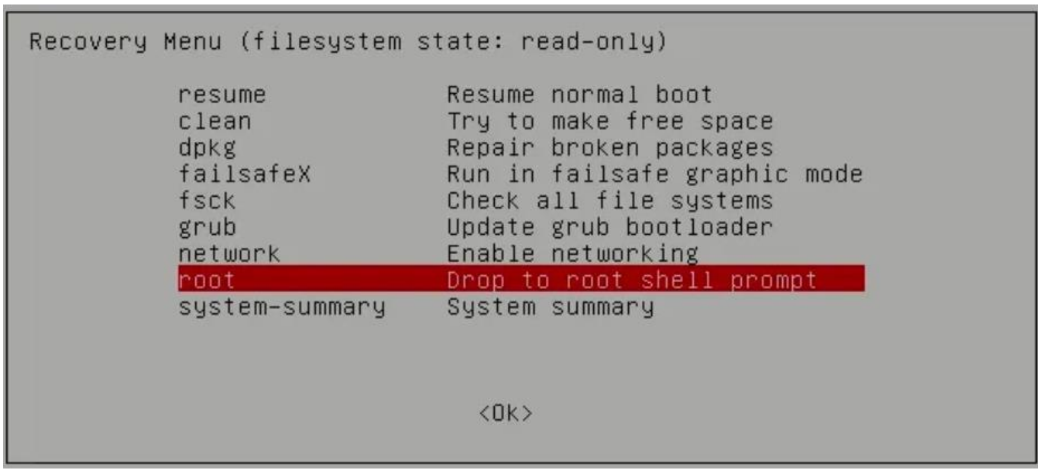


Source: It's foss

The **recovery Mode** option will appear. Press it to continue the process:

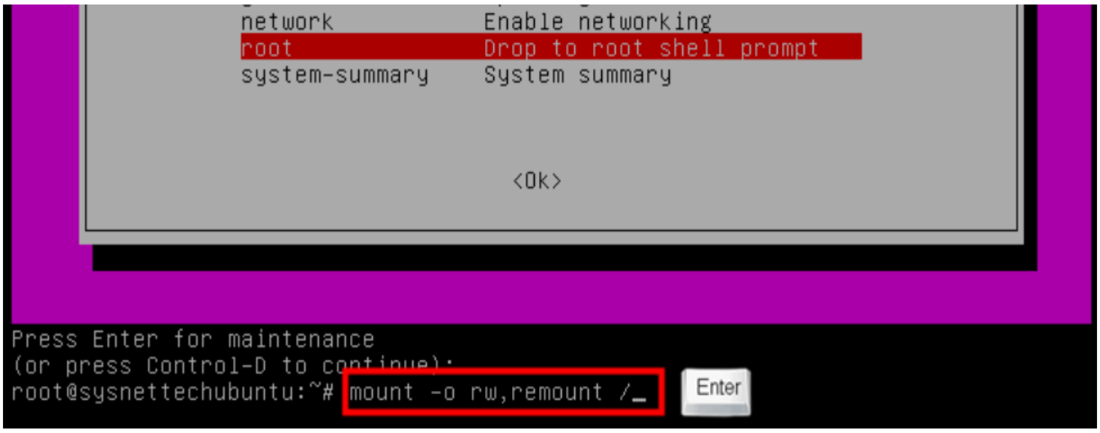
Source: It's foss

In **Recovery Mode** choose the option root-drop to root shell prompt:



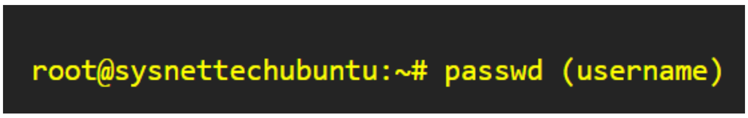
Source: It's foss

When Selectin the **root – drop to root shell prompt** option in recovery mode, the command line will be displayed look at image below. Soon after, it will be necessary to reassemble the read- only file system for write permission it important by executing the command below:



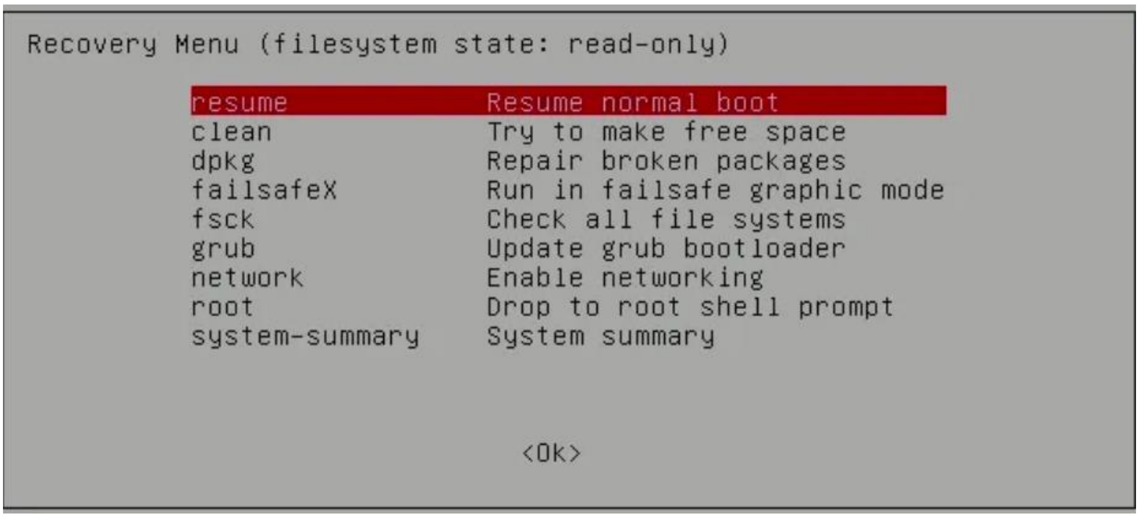
Source: It's foss

Ready! It is possible now to run the password recovery command look at how suppose to be shown on screen:



Source: It's foss

Set the new password and complete the operation by executing the exit command to return to the recovery menu, look at the image below. Then press the **Resume** option:



Source: It’s foss

When prompted for the user’s password on the login screen it must enter the new password registered in this operation!

Reference:

Prakash,A.,2021.How to Reset Ubuntu Password in 2 minutes .[online] available at: https://itsfoss.com/how-to-hack-ubuntu-password/ ,[Accessed 06 may 2021].